

3.

More notes on a, q, t -polynomials for generic curves.

3.1. Fix integers $M, N > 0$. Write $d = \gcd(M, N)$ and $M = md$ and $N = nd$.

We define a *lattice square* to be a unit square in the xy -plane whose vertices are lattice points. Given such a square $\square$, we define its *coordinates* $(x(\square), y(\square))$ to be the coordinates of its upper right corner. We define its *rank* to be

$$r(\square) = mnd - nx(\square) - my(\square).$$

Observe that $r(\square) \geq 0$ if and only if the lattice square lies below the line between $(M, 0)$ and $(0, N)$. Let

$$\mathcal{S}_{M,N} = \{\text{lattice squares } \square \mid 1 \leq y(\square) \leq N \text{ and } r(\square) \geq 0\}.$$

When $d = 1$, the elements of $\mathcal{S}_{M,N}$ are uniquely determined by their ranks, but in general, there can be up to d elements with the same rank.

3.2. We say that a lattice path is *monotone* when it only uses leftward and upward steps, and more precisely, *a, b -monotone* when it uses exactly a leftward steps and b upward steps. An *a, b -Dyck path* is an a, b -monotone lattice path that stays below the line connecting its starting and ending points. We often identify lattice paths with their images, viewed as closed subsets of $\mathbf{R}^2$.

Let $\mathcal{D}_{M,N}$ be the collection of M, N -Dyck paths that are contained in the region covered by $\mathcal{S}_{M,N}$: explicitly, $\mathcal{D}_{M,N} = \bigcup_{x \geq 0} \mathcal{D}_{M,N}^x$, where

$$\mathcal{D}_{M,N}^x = \{M, N\text{-Dyck paths starting at } (M - x, 0)\}.$$

Note that translation defines a bijection $\mathcal{D}_{M,N}^0 \xrightarrow{\sim} \mathcal{D}_{M,N}^x$ for each $x \geq 0$.

For any integers $a, b > 0$, let $\mathcal{I}_{a,b}$ be the collection of cofinite subsets $\Delta \subseteq \mathbf{Z}_{\geq 0}$ such that $\Delta + a \subseteq \Delta$ and $\Delta + b \subseteq \Delta$. For any $\Delta \in \mathcal{I}_{M,N}$ and residue j modulo N , let

$$\alpha_j(\Delta) = \min((j + N\mathbf{Z}) \cap \Delta).$$

We say that Δ is *admissible* if and only if, for every k modulo d , there is some j such that $\alpha_j(\Delta) \in k + d\mathbf{Z}$ and $\alpha_j(\Delta) + M + 1 \in \Delta$. Let $\mathcal{A}_{M,N} = \bigcup_{x \geq 0} \mathcal{A}_{M,N}^x$, where

$$\mathcal{A}_{M,N}^x = \{\text{admissible } \Delta \in \mathcal{I}_{M,N} \mid \min(\Delta) = x\}.$$

Again, translation defines a bijection $\mathcal{A}_{M,N}^0 \xrightarrow{\sim} \mathcal{A}_{M,N}^x$ for each $x \geq 0$.

The results of [GMV] produce an explicit bijection $\mathcal{A}_{M,N}^0 \xrightarrow{\sim} \mathcal{D}_{M,N}^0$, hence a bijection $\mathcal{A}_{M,N} \xrightarrow{\sim} \mathcal{D}_{M,N}$.

Example 3.1. We can check that $|\mathcal{D}_{6,4}^0| = |\mathcal{A}_{6,4}^0| = 23$.

Remark 3.2. Lemma 3.9 of [GMO] shows that if $\Delta \in \mathcal{I}_{M,N}$ is admissible, then $\Delta + mnd + 1 \subseteq \Delta$. The converse is false. For $M = 6$ and $N = 4$, we have $mnd + 1 = 13$. There are 25 elements $\Delta \in \mathcal{I}_{6,4}$ such that $0 \in \Delta$ and $\Delta + 13 \in \Delta$, two of which fail to be admissible: $\mathbf{Z}_{\geq 0} \setminus \{1, 3, 5, 7, 9, 11\}$ and $\mathbf{Z}_{\geq 0} \setminus \{1, 3, 5, 7, 9\}$. The corresponding vectors $(\alpha_0, \alpha_1, \alpha_2, \alpha_3)$ are $(0, 13, 2, 15)$ and $(0, 13, 2, 11)$.

3.3. It will be useful to reformulate admissibility. First, observe that a cofinite subset $\Delta \subseteq \mathbf{Z}_{\geq 0}$ can always be decomposed into

$$\Delta = \bigcup_{0 \leq k \leq d-1} (d\Theta_k(\Delta) + k)$$

for some cofinite subsets $\Theta_k(\Delta) \subseteq \mathbf{Z}_{\geq 0}$ uniquely determined by Δ . Note that if $\Delta \in \mathcal{J}_{M,N}$, then $\Theta_k(\Delta) \in \mathcal{J}_{m,n}$ for all k . It is explained in [GMO, §4.4] that:

Lemma 3.3. *An element $\Delta \in \mathcal{J}_{M,N}$ is admissible if and only if, for each $0 \leq k < d-1$, we have $\Theta_{k+1}(\Delta) \not\subseteq \Theta_k(\Delta) + m + n$.*

Example 3.4. Take $M = 6$ and $N = 4$. For $\Delta = \mathbf{Z}_{\geq 0} \setminus \{1, 3, 5, 7, 9\} \in \mathcal{J}_{6,4}$, we have $\Theta_0(\Delta) = \mathbf{Z}_{\geq 0}$ and $\Theta_1(\Delta) = \mathbf{Z}_{\geq 5}$. Thus, $\Theta_1(\Delta) = \Theta_0(\Delta) + 5$, confirming the observation from Remark 3.2 that Δ is not admissible.

For any cofinite subset $\Theta \subseteq \mathbf{Z}_{\geq 0}$, let

$$\begin{aligned} \text{cog}_m(\Theta) &= \{\max((i + m\mathbf{Z}) \setminus \Theta) \mid i = 0, 1, \dots, m-1\}, \\ \text{gen}_n(\Theta) &= \{\min((j + n\mathbf{Z}) \cap \Theta) \mid j = 0, 1, \dots, n-1\}. \end{aligned}$$

In terms of our earlier notation, $\{\alpha_j(\Delta)\}_j = \bigcup_k \text{gen}_n(\Theta_k(\Delta))$. Note that $\text{cog}_m(\Theta) \subseteq \mathbf{Z} \setminus \Theta$, whereas $\text{gen}_n(\Theta) \subseteq \Theta$. Let

$$s(\Theta) = \text{cog}_m(\Theta) \cup \text{gen}_n(\Theta).$$

In [GMV, GMO], $s(\Theta)$ is called the *skeleton* of Θ . Lemma 4.5 of [GMO] states:

Lemma 3.5. *If $\Delta \in \mathcal{J}_{M,N}$, then for any pair of distinct indices i, j , exactly one of the following must hold:*

- $s(\Theta_i(\Delta)) \cap s(\Theta_j(\Delta)) \neq \emptyset$.
- $\Theta_i(\Delta) \subseteq \Theta_j(\Delta) + m + n$.
- $\Theta_j(\Delta) \subseteq \Theta_i(\Delta) + m + n$.

Corollary 3.6. *An element $\Delta \in \mathcal{J}_{M,N}$ is admissible if and only if, for each $0 \leq k < d-1$, we have either $s(\Theta_k(\Delta)) \cap s(\Theta_{k+1}(\Delta)) \neq \emptyset$ or $\Theta_k(\Delta) \subseteq \Theta_{k+1}(\Delta) + m + n$.*

3.4. To describe the bijection $\mathcal{D}_{M,N} \xrightarrow{\sim} \mathcal{A}_{M,N}$, we need several preliminaries.

3.4.1. For any lattice square $\square$, let $e_{\downarrow}(\square)$ and $e_{\rightarrow}(\square)$ denote the bottom and right-hand edges of $\square$, respectively. For $* \in \{\downarrow, \rightarrow\}$, and for any monotone lattice path P , let

$$\begin{aligned} s_*(P) &= \{\text{lattice squares } \square \mid e_*(\square) \subseteq P\}, \\ r_*(P) &= \{r(\square) \mid \square \in s_*(P)\}. \end{aligned}$$

Note that if P is m, n -monotone, then it can be recovered up to translation by $(-m, n)$ from either $r_{\downarrow}(P)$ or $r_{\rightarrow}(P)$.

In particular, if $d = 1$, then any Dyck path $D \in \mathcal{D}_{M,N}$ can be recovered from either $r_{\downarrow}(D)$ or $r_{\rightarrow}(D)$. Here, the bijection $\mathcal{D}_{M,N} \xrightarrow{\sim} \mathcal{A}_{M,N}$ can be characterized by either

of the equalities of sets $r_{\downarrow}(D) = \text{cog}_n(\Delta)$ or $r_{\rightarrow}(D) = \text{gen}_n(\Delta)$. If instead $d > 1$, then there can be multiple elements of $s_{\downarrow}(D)$ or $s_{\rightarrow}(D)$ with the same rank. Even when this does not occur, neither $r_{\downarrow}(D)$ nor $r_{\rightarrow}(D)$ need determine D .

It will be convenient to have a refinement of $r_{\rightarrow}(D)$ that does determine D . Let $\square_i(D)$ be the unique element of $s_{\rightarrow}(D)$ with $y(\square_i(D)) = i$. Then D can be recovered from

$$\rho(D) = (r(\square_1(D)), \dots, r(\square_N(D))).$$

Note that $r_{\rightarrow}(D)$ is just the set of values attained by the n -tuple $\rho(D)$.

Example 3.7. There are two distinct elements $D \in \mathcal{D}_{6,4}$ such that the ranks $r(\square)$ for $\square \in s_{\rightarrow}(D)$ are precisely 0, 1, 2, 3, each occurring once.

Figure 10.10 shows two 7x7 payoff matrices. The left matrix has a saddle point at (3,3) with a payoff of 3. The right matrix has a saddle point at (3,4) with a payoff of 1. Both matrices have a diagonal line from (0,6) to (6,0).

The respective sequences $\rho(D)$ are $(3, 2, 1, 0)$ and $(1, 2, 3, 0)$.

3.4.2. For any monotone lattice path P , we set

$$\begin{aligned}s(P) &= s_{\downarrow}(P) \cup s_{\rightarrow}(P), \\ r(P) &= r_{\downarrow}(P) \cup r_{\rightarrow}(P).\end{aligned}$$

Again following [GMO, GMV], we refer to $r(P)$ as the *skeleton* of P .

Given $D \in \mathcal{D}_{M,N}$, a *removable segment* of D is a lattice sub-path $P \subseteq D$ such that:

- P is m, n -monotone.
- $s(P)$ contains the rightmost (or bottommost) lattice square $\square$ such that $r(\square) \in r(D) \cap r(P)$.

For such P , there is a sequence of n consecutive indices i such that $e_{\rightarrow}(\square_i(D)) \subseteq P$. These indices cut out a subsequence $\rho(P)$ of the tuple $\rho(D)$. We set $P_{d-1} = P$. We delete $\rho(P)$ from $\rho(D)$ to obtain the tuple $\rho(D_1)$ for some $D_1 \in \mathcal{D}_{M-m, N-n}$. We can then look for a removable segment P_{d-2} of D_1 , delete the corresponding subsequence $\rho(P_{d-2})$ of $\rho(D_1)$ to find $D_2 \in \mathcal{D}_{M-2m, N-2n}$, *etc.* Repeating this process, we obtain a list of m, n -monotone lattice paths

$$P_{d-1}, P_{d-2}, \dots, P_1, P_0.$$

Note that the order of this list is not uniquely determined by D , because at any given step of the process, there could be multiple removable segments. However, the set $\{P_i\}_i$ is determined by D .

In fact, more is true. Define a partial order $<$ on the set $\{P_i\}_i$ as follows: $i < j$ if and only if $r(P_i) \cap r(P_j) \neq \emptyset$ and P_i can be removed *after* P_j is removed in the process

above. Let $\Pi(D)$ denote the resulting poset. It is shown in [GMV] that $\Pi(D)$ is uniquely determined by D .

We define a *minimal shift* for D to be a bijection from $\{0, 1, \dots, d-1\}$ onto $\Pi(D)$, say $k \mapsto Q_k$, such that $k < l$ whenever $Q_k < Q_l$. For such Q and $0 \leq k \leq d-1$, let $\Theta_k(D, Q)$ be the unique element of $\mathcal{J}_{m,n}$ with skeleton Q_k . Let $\Delta(D, Q) \in \mathcal{J}_{M,n}$ be defined by

$$\Delta(D, Q) = \bigcup_{0 \leq k \leq d-1} (d\Theta_k(D, Q) + k).$$

The main combinatorial result of [GMO] is:

Theorem 3.8. *For any $D \in \mathcal{D}_{M,N}^0$, there is a unique minimal shift $Q^\star$ such that, writing $\Theta_k(D) := \Theta_k(D, Q^\star)$, we have $\Theta_k(D) \subseteq \Theta_{k+1}(D) + m + n$ whenever $Q_k^\star \not\prec Q_{k+1}^\star$. That is, there is a unique minimal shift $Q^\star$ such that $\Delta(D) := \Delta(D, Q^\star)$ is admissible.*

Moreover, every element $\Delta \in \mathcal{A}_{M,N}^0$ arises this way from a unique D . That is, the map

$$D \mapsto \Delta(D) : \mathcal{D}_{M,N}^0 \rightarrow \mathcal{A}_{M,N}^0$$

is a bijection.

Example 3.9. Take $M = 8$ and $N = 12$, so that $d = 4$ and $m = 2$ and $n = 3$. Consider the following Dyck path $D \in \mathcal{D}_{8,12}$:



We can describe the list $P_3, \dots, P_0$ in terms of skeletons:

$$\begin{aligned} s_{\downarrow}(P_3) &= \{-2, -1\}, & s_{\rightarrow}(P_3) &= \{0, 2, 1\}, \\ s_{\downarrow}(P_2) &= \{4, 5\}, & s_{\rightarrow}(P_2) &= \{6, 8, 7\}, \\ s_{\downarrow}(P_1) &= \{4, 7\}, & s_{\rightarrow}(P_1) &= \{6, 8, 10\}, \\ s_{\downarrow}(P_0) &= \{-2, 1\}, & s_{\rightarrow}(P_0) &= \{0, 2, 4\}. \end{aligned}$$

The poset $\Pi(D)$ is generated by the relations $P_0 < P_1 < P_2$ and $P_0 < P_3$. It turns out that in this example, $Q_i^\star = P_i$ for all i . Explicitly,

$$\Theta_0(D) = \mathbf{Z}_{\geq 0} \setminus \{1\},$$

$$\Theta_1(D) = \mathbf{Z}_{\geq 6} \setminus \{7\},$$

$$\Theta_2(D) = \mathbf{Z}_{\geq 6},$$

$$\Theta_3(D) = \mathbf{Z}_{\geq 0}.$$

Admissibility amounts to the fact that $\Theta_2(D) \subseteq \mathbf{Z}_{\geq 5} = \Theta_3 + 5$.

3.5. We are interested in how the bijection $\mathcal{D}_{M,N} \xrightarrow{\sim} \mathcal{A}_{M,N}$ interacts with certain numerical invariants.

3.5.1. For any lattice square $\square$, let $e_\uparrow(\square)$ denote the upper edge of $\square$. For any Dyck path $D \in \mathcal{D}_{M,N}$, let

$$\mathcal{S}(D) = \{\square \in \mathcal{S}_{M,N} \mid (r(\square) < r(\square_i(D)), \text{ where } i = y(\square)),$$

$$s_{\nearrow}(D) = \{\square \in \mathcal{S}_{M,N} \mid e_{\rightarrow}(\square), e_{\uparrow}(\square) \subseteq D\} \cup \{\square_N(D)\}.$$

Using $\mathcal{S}(D)$, we define

$$h(D) = |\{(\square, \square') \in \mathcal{S}(D) \times s_{\rightarrow}(D) \mid r(\square') < r(\square) \leq r(\square') + m\}|.$$

For all $\square \in s_{\nearrow}(D)$, we define

$$g(D, \square) = |\{\square' \in s_{\rightarrow}(D) \mid r(\square') \leq r(\square) < r(\square') + m\}|.$$

Nota bene the distinction between the $\leq$'s and the $<$'s. Also note that we always have $g(D, \square_N(D)) = 0$.

3.5.2. For all $a \in \mathbf{Z}$ and $\Delta \in \mathcal{S}_{M,N}$, let

$$\dim(\Delta, \Delta') = \sum_{\substack{j \\ \alpha_j(\Delta) \notin \Delta'}} |\mathbf{Z}_{\geq \alpha_j(\Delta)} \setminus \Delta| + \sum_{\substack{j \\ \alpha_j(\Delta) \in \Delta'}} |\mathbf{Z}_{\geq \alpha_j(\Delta)} \setminus \Delta'| - \sum_j |\mathbf{Z}_{\geq \alpha_j(\Delta)+M} \setminus \Delta'|,$$

$$\dim(\Delta) = \dim(\Delta, \Delta)$$

$$= \sum_j |\mathbf{Z}_{\geq \alpha_j(\Delta)} \setminus \Delta| - \sum_j |\mathbf{Z}_{\geq \alpha_j(\Delta)+M} \setminus \Delta|.$$

The following result should be in [GMV], once we rewrite h in terms of “ din ”.

Theorem 3.10 (GMV). *If $D \in \mathcal{D}_{M,N}$ maps to $\Delta \in \mathcal{A}_{M,N}$, then*

$$h(D) = \dim(\Delta).$$

3.5.3. For any $\Delta \in \mathcal{A}_{M,N}$, let

$$\text{gen}(\Delta) = \{a \in \Delta \mid a - M, a - N, a - M - 1 \notin \Delta\}.$$

Let $\Delta_+ = \Delta \setminus \text{gen}(\Delta)$. Equivalently,

$$\Delta_+ = (\Delta + M) \cup (\Delta + N) \cup ((\Delta + M + 1) \cap \Delta).$$

We claim the further compatibility:

Theorem 3.11. *If $D \in \mathcal{D}_{M,N}$ maps to $\Delta \in \mathcal{A}_{M,N}$, then*

$$(3.1) \quad t^{h(D)} \prod_{\square \in s_{\nearrow}(D)} (1 + at^{g(D, \square)}) = \sum_{l \geq 0} a^l t^{\binom{l}{2}} \sum_{\substack{\Delta' \in \mathcal{A}_{M,N} \\ \Delta \supseteq \Delta' \supseteq \Delta_+ \\ |\Delta \setminus \Delta'| = l}} t^{\dim(\Delta, \Delta')}.$$

Note that since $g(D, \square_N(D)) = 0$, the product on the left-hand side is always divisible by $1 + a$. Also note that setting $a = 0$ recovers the identity $h(D) = \dim(\Delta)$.

Example 3.12. The simplest nontrivial case: $M = 3$ and $N = 2$. Here, $d = 1$, so the bijection $\mathcal{D}_{3,2} \xrightarrow{\sim} \mathcal{A}_{3,2}$ is characterized by $r_{\rightarrow}(D) = \{\alpha_0(\Delta), \alpha_1(\Delta)\}$. The elements of $\mathcal{A}_{3,2}^0$ are $\mathbf{Z}_{\geq 0}$ and $\mathbf{Z}_{\geq 0} \setminus \{1\}$. The corresponding tuples $(\alpha_0(\Delta), \alpha_1(\Delta))$ are $(0, 1)$ and $(0, 3)$, respectively.

- For $\Delta = \mathbf{Z}_{\geq 0}$, we have $\Delta_+ = \mathbf{Z}_{\geq 2}$. Here, (3.1) says

$$1 \cdot (1 + a) \cdot (1 + at) = 1 \cdot 1 + a \cdot (1 + t) + a^2 t \cdot 1,$$

as the possibilities for Δ' are $\mathbf{Z}_{\geq 0}, \mathbf{Z}_{\geq 0} \setminus \{1\}, \mathbf{Z}_{\geq 1}, \mathbf{Z}_{\geq 2}$.

- For $\Delta = \mathbf{Z}_{\geq 0} \setminus \{1\}$, we again have $\Delta_+ = \mathbf{Z}_{\geq 2}$. Here, (3.1) says

$$t \cdot (1 + a) = 1 \cdot t + a \cdot t,$$

as the possibilities for Δ' are $\mathbf{Z}_{\geq 0} \setminus \{1\}$ and $\mathbf{Z}_{\geq 2}$.

Example 3.13. Take $M = 6$ and $N = 4$. It turns out that there is a unique Dyck path $D \in \mathcal{D}_{6,4}^0$ such that $r_{\rightarrow}(D) = \{0, 1, 4, 5\}$. Writing $s_{\rightarrow}(D) = \{\square_1, \square_2, \square_3, \square_4\}$, where $\square_i = \square_i(D)$ in our earlier notation, we get $s_{\nearrow}(D) = \{\square_1, \square_3, \square_4\}$. We have:

$$h(D) = 5, \quad g(D, \square_1) = g(D, \square_3) = 1, \quad g(D, \square_4) = 0.$$

The element $\Delta \in \mathcal{A}_{6,4}^0$ corresponding to D must have $(\alpha_j(\Delta))_j = (0, 9, 2, 11)$, from which $\Delta = \mathbf{Z}_{\geq 0} \setminus \{1, 3, 5, 7\}$. We get $\text{gen}(\Delta) = \{0, 2, 11\}$, from which $(\alpha_j(\Delta'))_j = (4, 9, 6, 15)$, from which $\Delta_+ = \mathbf{Z}_{\geq 4} \setminus \{5, 7, 11\}$. The elements $\Delta' \in \mathcal{A}_{6,4}$ with $\Delta \supseteq \Delta' \supseteq \Delta_+$ are:

$ \Delta \setminus \Delta' $	$(\alpha_j(\Delta'))_j$
0	$(0, 9, 2, 11)$
1	$(0, 9, 2, 15), (0, 9, 6, 11), (4, 9, 2, 11)$
2	$(0, 9, 6, 15), (4, 9, 2, 15), (4, 9, 6, 11),$
3	$(4, 9, 6, 15)$

All of these elements Δ' are admissible. One can check (3.1) by hand.

3.6. We rewrite the two sides of (3.1) to simplify the comparison.

3.6.1. First, we see that

$$\begin{aligned} \prod_{\square \in s_{\nearrow}(D)} (1 + at^{g(D, \square)}) &= \sum_{s' \subseteq s_{\nearrow}(D)} a^{|s'|} t^{\sum_{\square \in s'} g(D, \square)} \\ &= \sum_{l \geq 0} a^l \sum_{\substack{s' \subseteq s_{\nearrow}(D) \\ |s'|=l}} t^{\sum_{\square \in s'} g(D, \square)}. \end{aligned}$$

3.6.2. For any $\Delta, \Delta' \in \mathcal{A}_{M, N}$ with $\Delta \supseteq \Delta' \supseteq \Delta_+$, we compute:

$$\begin{aligned} \dim(\Delta, \Delta') - \dim(\Delta) &= \sum_{\substack{j \\ \alpha_j(\Delta) \in \Delta'}} (|\mathbf{Z}_{\geq \alpha_j(\Delta)} \setminus \Delta'| - |\mathbf{Z}_{\geq \alpha_j(\Delta)} \setminus \Delta|) \\ &\quad - \sum_j (|\mathbf{Z}_{\geq \alpha_j(\Delta)+M} \setminus \Delta'| - |\mathbf{Z}_{\geq \alpha_j(\Delta)+M} \setminus \Delta|) \\ &= \sum_{\substack{j \\ \alpha_j(\Delta) \in \Delta'}} |\mathbf{Z}_{\geq \alpha_j(\Delta)} \cap (\Delta \setminus \Delta')| - \sum_j |\mathbf{Z}_{\geq \alpha_j(\Delta)+M} \cap (\Delta \setminus \Delta')| \\ &= \sum_{\substack{j \\ \alpha_j(\Delta) \in \Delta'}} |[\alpha_j(\Delta), \alpha_j(\Delta) + M) \cap (\Delta \setminus \Delta')| \\ &\quad - \sum_{\substack{j \\ \alpha_j(\Delta) \notin \Delta'}} |\mathbf{Z}_{\geq \alpha_j(\Delta)+M} \cap (\Delta \setminus \Delta')|. \end{aligned}$$

So if $|\Delta \setminus \Delta'| = l$, then

$$\begin{aligned} &\binom{l}{2} + \dim(\Delta, \Delta') - \dim(\Delta) \\ &= \binom{l}{2} - \sum_{\substack{j \\ \alpha_j(\Delta) \notin \Delta'}} |\mathbf{Z}_{\geq \alpha_j(\Delta)+M} \cap (\Delta \setminus \Delta')| + \sum_{\substack{j \\ \alpha_j(\Delta) \in \Delta'}} |[\alpha_j(\Delta), \alpha_j(\Delta) + M) \cap (\Delta \setminus \Delta')| \\ &= |\{(a, b) \in (\Delta \setminus \Delta')^2 \mid a < b < a + M\}| + \sum_{\substack{j \\ \alpha_j(\Delta) \in \Delta'}} |(\alpha_j(\Delta), \alpha_j(\Delta) + M) \cap (\Delta \setminus \Delta')| \\ &= \sum_j |(\alpha_j(\Delta), \alpha_j(\Delta) + M) \cap (\Delta \setminus \Delta')|. \end{aligned}$$

To get the second equality above, we have used the fact that $\Delta \setminus \Delta' \subseteq \text{gen}(\Delta) \setminus \Delta' \subseteq \{\alpha_j(\Delta)\}_j \setminus \Delta' \subseteq \Delta \setminus \Delta'$, from which $\Delta \setminus \Delta' = \{\alpha_j(\Delta)\}_j \setminus \Delta'$. We are left with

$$\binom{l}{2} + \dim(\Delta, \Delta') - \dim(\Delta) = \sum_j |(\alpha_j(\Delta), \alpha_j(\Delta) + M) \cap (\Delta \setminus \Delta')|.$$

3.6.3. The last two displays suggest looking for a bijection from the collection of size- l subsets $s' \subseteq s_{\nearrow}(D)$ to the collection of $\Delta' \in \mathcal{A}_{M, N}$ such that $\Delta \supseteq \Delta' \supseteq \Delta_+$ and $|\Delta \setminus \Delta'| = l$, such that $\sum_{\square \in s'} g(D, \square)$ is given by the last expression above.

First, we claim that the construction of the bijection $D \mapsto \Delta : \mathcal{D}_{M,N} \xrightarrow{\sim} \mathcal{A}_{M,N}$ implicitly provides a bijection

$$s_{\rightarrow}(D) \xrightarrow{\sim} \{\alpha_j(\Delta)\}_j.$$

Namely, given $\square \in s_{\rightarrow}(D)$, let $k(\square) \in \{0, 1, \dots, d-1\}$ be the integer such that $e_{\rightarrow}(\square) \subseteq Q_{k(\square)}^*$, where Q^* is the minimal shift appearing in Theorem 3.8. Then $r(\square) \in \text{gen}_n(\Theta_{k(\square)}(D))$, so $dr(\square) + k(\square)$ takes the form $\alpha_{j(\square)}(\Delta)$ for some $j(\square)$. The bijection sends

$$\square \mapsto dr(\square) + k(\square) = \alpha_{j(\square)}(\Delta).$$

Lemma 3.14. *When $D \mapsto \Delta$, the above bijection $s_{\rightarrow}(D) \xrightarrow{\sim} \{\alpha_j(\Delta)\}_j$ restricts to a further bijection $s_{\nearrow}(D) \xrightarrow{\sim} \text{gen}(\Delta)$.*

Proof. **TODO** □

Next, observe that there is a bijection between subsets of $\Gamma \subseteq \text{gen}(\Delta)$ and sets Δ' such that $\Delta \supseteq \Delta' \supseteq \Delta_+$: namely, $\Delta' = \Delta \setminus \Gamma$. Any set Δ' arising this way is indeed an element of $\mathcal{A}_{M,N}$. So we arrive at a bijection

$$s' \mapsto \Delta(s') : \{\text{size-}l \text{ subsets } s' \subseteq s_{\nearrow}(D)\} \xrightarrow{\sim} \{\Delta' \in \mathcal{A}_{M,N} \mid \Delta \supseteq \Delta' \supseteq \Delta_+, |\Delta \setminus \Delta'| = l\}.$$

Now the following result finishes the proof of Theorem 3.11.

Lemma 3.15. *If s' maps to Δ' above, then*

$$\sum_{\square \in s'} g(D, \square) = \sum_j |(\alpha_j(\Delta), \alpha_j(\Delta) + M) \cap (\Delta \setminus \Delta')|.$$

Proof. We have

$$\begin{aligned} & \sum_{\square \in s'} g(D, \square) \\ &= |\{(\square, \square') \in s' \times s_{\rightarrow}(D) \mid r(\square') \leq r(\square) < r(\square') + m\}| \\ &= |\{(\square, \square') \in s' \times s_{\rightarrow}(D) \mid \alpha_{j(\square')}(\Delta) \leq dr(\square) + k(\square') < \alpha_{j(\square')}(\Delta) + M\}| \end{aligned}$$

□